# Introduction to Quantitative Biology

# Lecture Notes: Inflammation and Fibrosis as a Bistable System

From Systems Medicine:
Physiological Circuits and the Dynamics of Disease
– Uri Alon

November 2026

This lecture note follows Chapter 5 of Uri Alon's *Systems Medicine: Physiological Circuits and the Dynamics of Disease* [1], aiming to understand the essence of inflammation and fibrosis. The central question is how a single biological process, tissue repair, can lead to two very different results: healing or fibrosis.

# 1. Inflammation and Fibrosis as a Bistable System

## 1.1 Inflammation Can Lead to Either Healing or Fibrosis

**Fibrosis** is excessive scarring that occurs in the liver, lung, kidney, heart, and other organs, and is a major contributor to age-related diseases. In fibrosis, scar tissue replaces healthy tissue and the organ loses function. Fibrosis is always preceded by periods of intense **inflammation**. There is a universal sequence across tissues:

$$\text{injury} \longrightarrow \text{inflammation} \longrightarrow \text{healing or fibrosis}$$

The process of tissue repair can lead to two different outcomes, depending on the duration and intensity of the injury. In organs with poor ability to regenerate, like the heart, fibrosis is triggered by almost any injury. In organs that can repair themselves to a certain extent, transient or small injuries lead to healing, but prolonged, repetitive, or extensive injuries cause fibrosis.

We will take a big-picture view, putting the essential facts of inflammation and fibrosis into a mathematical model that captures the core features. This model has a basic property called **multistability** – the ability to produce two or more different stable steady states. Multistability can shed light on how inflammation can lead either to healing or to fibrosis.

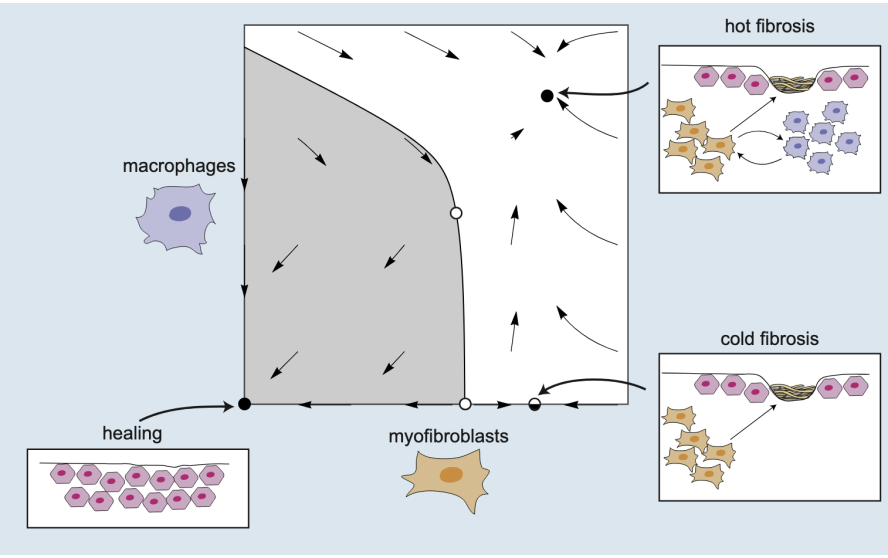



Figure 1: Inflammation can lead to either healing or fibrosis, showing the property of multistability.

## 1.2 Inflammation Includes Influx of Macrophages and Activation of Myofibroblasts

Injury to a tissue causes cells to release factors that cause inflammation. The purpose of inflammation is to fight pathogens and to start repair. Inflammation is characterized by the following features: redness, heat, swelling, and pain, together with loss of tissue function.

Redness, swelling, and heat are caused by the dilation of nearby blood vessels, which open up to let immune cells flow into the tissue, together with fluids and proteins that fight pathogens. The main immune cells are white blood cells. Among them are blood monocytes that turn into **macrophages**, cells that can engulf pathogens and dead cells. Macrophages play a big role in fibrosis and healing.

In parallel, damaged cells and incoming macrophages secrete signals to activate fibroblasts. These tissue-resident fibroblasts proliferate and differentiate into **myofibroblasts**. Their muscle-like ability to generate force helps to contract and close the wound.

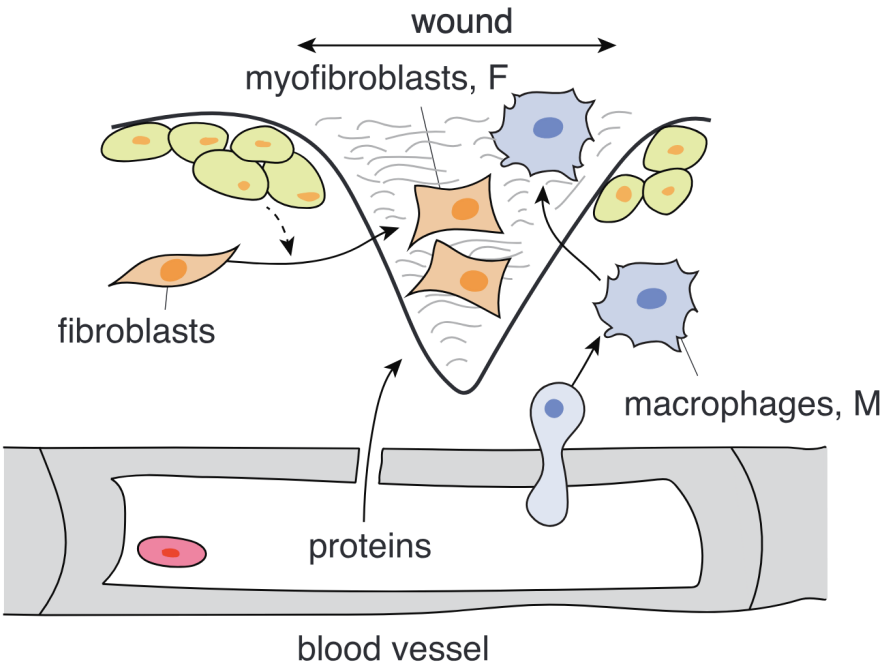



Figure 2: Inflammation at an injury site involves influx of fluids and macrophages together with differentiation of fibroblasts into myofibroblasts.

The two main cell types are thus the incoming macrophages, denoted $M$, and the myofibroblasts, denoted $F$. These two cell types activate each other's proliferation by secreting growth factors for each other – small proteins that diffuse in the tissue and are sensed by receptors on the cell surface. F cells also secrete a growth factor for themselves, in an example of an **autocrine loop**.

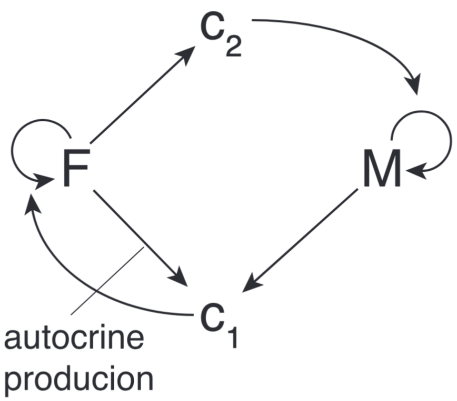



Figure 3: Macrophages $M$ and myofibroblasts $F$ secrete growth factors $c_1$ and $c_2$ for each other, and myofibroblasts also secrete growth factors for themselves in an autocrine loop.

Our purpose is to understand the dynamics of the inflammation process and how it can "decide" to show healing or fibrosis.

# 2. Mathematical Model for Myofibroblasts Shows Bistability

We begin by considering only the myofibroblasts, $F$. The main point is that one equation for myofibroblasts can show two different behaviors, a property called bistability.

The myofibroblasts produce and secrete a growth factor for themselves, denoted $c_1$. If each $F$ cell secretes it at rate $a$ and it is degraded at rate $\gamma_1$, its rate of change is production minus removal

$$\frac{dc_1}{dt} = aF - \gamma_1 c_1 \tag{2.1}$$

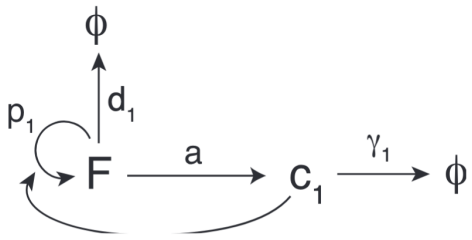



Figure 4: Circuit describing how myofibroblasts secrete their own growth factor and are removed.

Growth-factor production and removal take minutes to hours, whereas cell division and death take a day or longer. On the rapid timescale of $c_1$, cell density $F$ hardly changes. Thus, growth factors like $c_1$ are in quasi-steady-state, which we can compute by setting $dc_1/dt = 0$. We find that $c_1$ is proportional to the cells that make it

$$c_1 = \frac{aF}{\gamma_1} \tag{2.2}$$

The rate of change of $F$ is cell proliferation minus cell removal. Each cell proliferates at rate $p_1 c_1$ and is removed at rate $d_1$,

$$\frac{dF}{dt} = p_1 c_1 F - d_1 F \tag{2.3}$$

Substituting the quasi-steady-state value of $c_1$ results in a proliferation that rises as $F^2$,

$$\frac{dF}{dt} = \frac{p_1 a}{\gamma_1} F^2 - d_1 F \tag{2.4}$$

We then find the fixed points where $dF/dt = 0$, and use the rate plot. The total cell proliferation $\frac{p_1 a}{\gamma_1} F^2$ rises quadratically with $F$, since they make their own growth factor. And the

total cell removal $d_1 F$ is a line that rises with $F$. The curves intersect at the fixed points: a stable fixed point at zero and a second unstable fixed point at $F_u$.

$$F\left(\frac{p_1 a}{\gamma_1}F - d_1\right) = 0 \quad \Longrightarrow \quad F = 0, \quad F_u = \frac{d_1\gamma_1}{p_1 a} \tag{2.5}$$

For $0 < F < F_u$, the removal line lies above the proliferation curve, so $F$ decreases toward zero. For $F > F_u$, proliferation exceeds removal and $F$ rises to infinity, spiraling out of control.

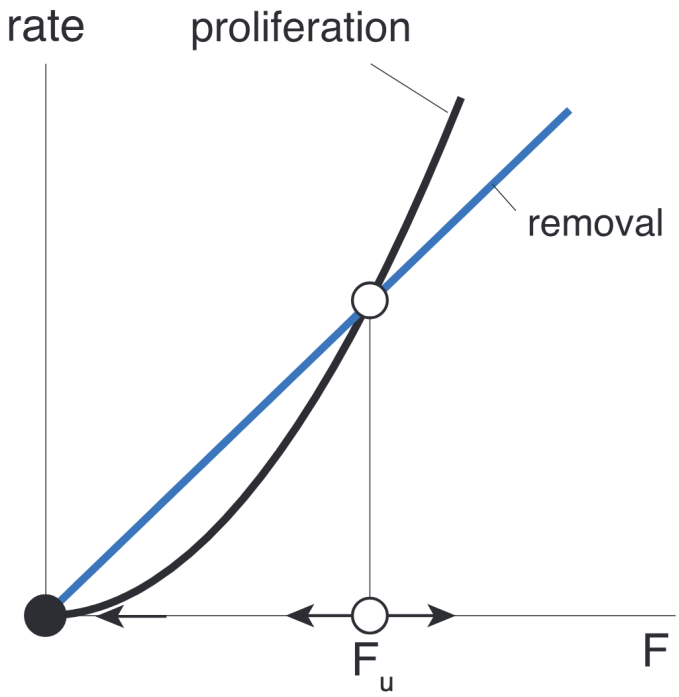



Figure 5: Rate plot of total cell proliferation and removal

Such a rise to infinity is not biologically feasible. Fibroblasts can sense the density of other fibroblasts and stop growing when they are too dense. The density limit is called a **carrying capacity**, denoted $K$. Carrying capacity reduces the proliferation rate by the term $1 - F/K$, which is close to one at low density and goes to zero at $F = K$. Such a linear reduction term for growth rate is called **logistic growth**, and is observed experimentally in fibroblasts in vitro. The equation with carrying capacity now reads

$$\frac{dF}{dt} = \frac{p_1 a}{\gamma_1}F^2\left(1 - \frac{F}{K}\right) - d_1 F \tag{2.6}$$

We find the fixed points using a new rate plot. Now the total proliferation now falls to zero at $F = K$. If removal rate $d_1$ is not too large, the removal curve crosses the proliferation curve three times: at zero, at an intermediate concentration, and at a high concentration of cells $F_{\text{high}}$.

The middle-fixed point is an unstable fixed point, $F_u$. If $F$ is smaller than $F_u$, the proliferation curve is lower than death and thus $F$ flows to zero. If $F$ is larger than $F_u$, F flows to the high fixed point, $F_{\text{high}}$. Thus $F = 0$ and $F = F_{\text{high}}$ are two stable fixed points. This feature, two stable fixed points for the same equation, is called **bistability**. Depending on initial conditions, the system flows to one of two possible stable states.

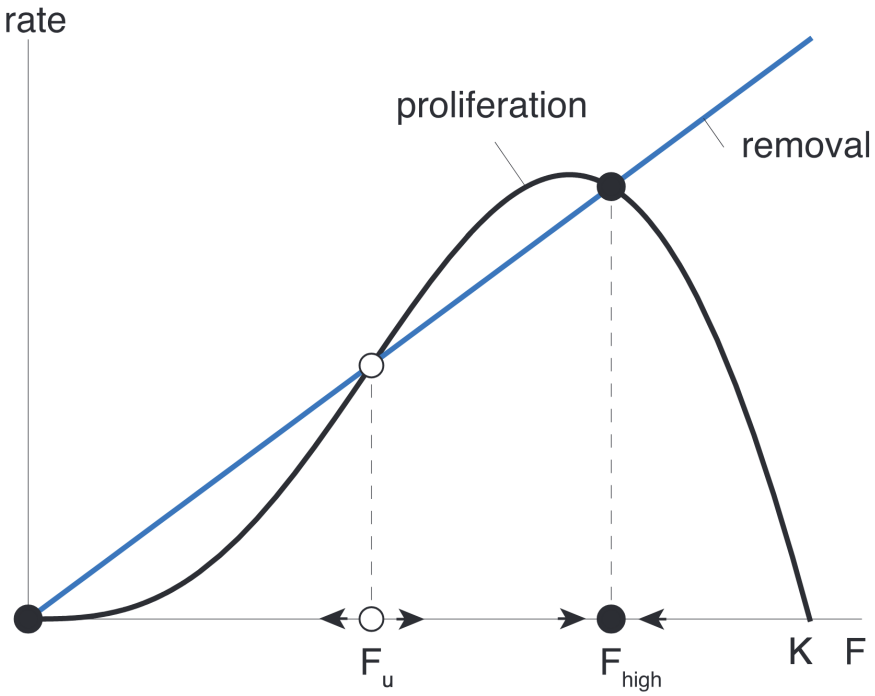



Figure 6: With carrying capacity, growth and removal curves can cross three times, generating two stable fixed points with an intermediate unstable fixed point.

If the death rate is too high, or proliferation is too low, the removal and production curves cross only once. A change of parameters causes the loss of a stable fixed point. This loss of bistability occurs when parameters that remove $F$ cells exceed parameters that favor $F$ cells.

To find the condition for bistability in the model for fibroblasts, Eq. (2.6), we set $dF/dt = 0$ to solve for the nonzero fixed points. This gives

$$d_1 = \frac{p_1 a}{\gamma_1} F \left(1 - \frac{F}{K}\right)$$

Then divide by $K$ and rearrange the parameters

$$\begin{aligned} \frac{\gamma_1 d_1}{p_1 a K} &= \frac{F}{K}\left(1 - \frac{F}{K}\right) \\ &= \frac{1}{4} - \left(\frac{F}{K} - \frac{1}{2}\right)^2 \end{aligned}$$

The right-hand side is a symmetric parabola that is zero at $F = 0$ and $F = K$. Its maximum value occurs between the two roots at $F = K/2$, where it reaches $1/4$. The horizontal line on the left crosses this parabola twice when it is less than $1/4$, giving the condition for bistability

$$\frac{p_1 aK}{\gamma_1 d_1} > 4 \tag{2.7}$$

When $p_1 aK/(\gamma_1 d_1) < 4$, the line lies above the parabola and only the stable zero state remains. This threshold compares parameters favoring $F$ – the carrying capacity $K$, autocrine growth factor secretion $a$ and $F$ proliferation rate $p_1$, relative to parameters opposing $F$ such as death rate $d_1$ or growth-factor removal rate $\gamma_1$. Exactly at the threshold, the removal curve touches the proliferation curve at a half-stable point, in addition to the zero fixed point.

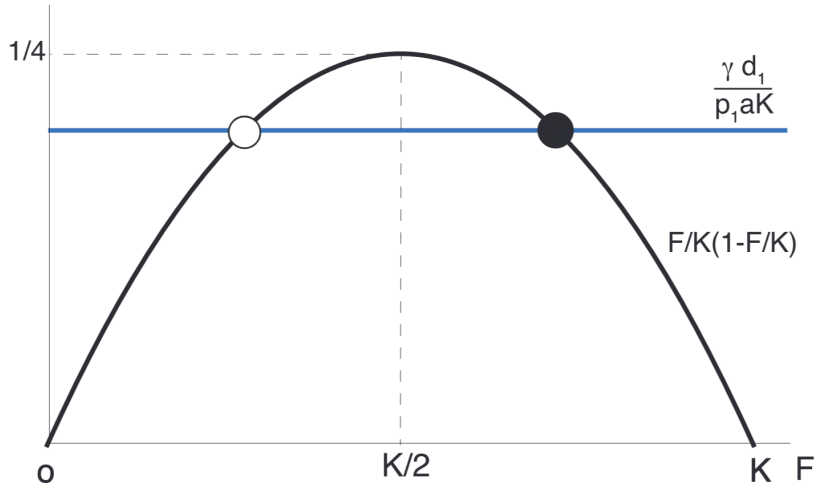



Figure 7: Rate plot for the growth and removal rates of fibroblasts shows how the number of fixed points depends on parameters.

# 3. The Macrophage–Myofibroblast Circuit Provides Two Fibrosis States and a Healing State

We now add macrophages $M$. The two cell-types together form a circuit that generates multistability.

## 3.1 Endocytosis Stabilizes the Two-Cell Circuit

The $F$ cells enhance proliferation of macrophage by secreting an $M$-specific growth factor $c_2$. Macrophages $M$ support $F$ proliferation by secreting an $F$-specific growth factor $c_1$. The myofibroblasts $F$ also produce $c_1$ for themselves as described above.

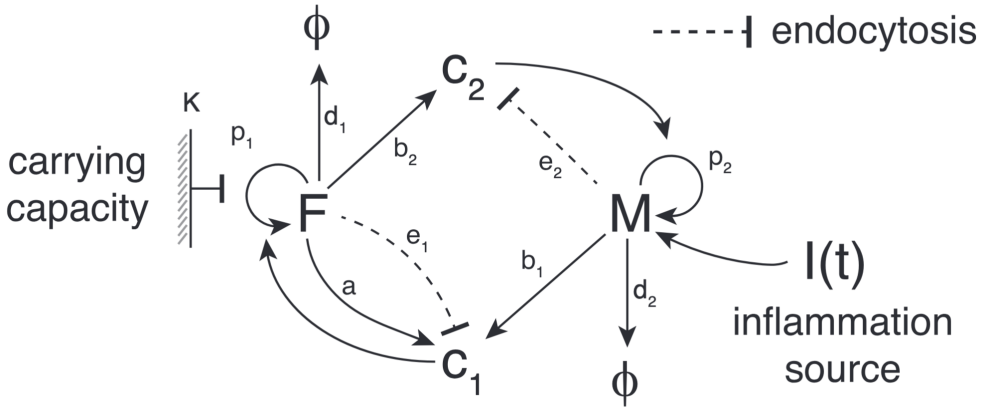



Figure 8: The macrophage-fibroblast circuit in which both cell types secrete growth factors for each other and deplete the growth factors by endocytosis. Fibroblasts have a carrying capacity and secrete their own growth factors in an autocrine loop.

Unlike $F$ cells, the $M$ cells have a very high carrying capacity: their numbers can increase by tens of folds when inflammation causes a large influx. However, they rarely approach their very high-carrying capacity due to a negative feedback mechanism: $M$ cells suck in the receptor on their membrane when it binds to the growth factor, degrading the growth factor and sometimes the receptor too – a process called **endocytosis**. Thus, excess $M$ cells consume their own growth factor, reducing their numbers back to steady-state.

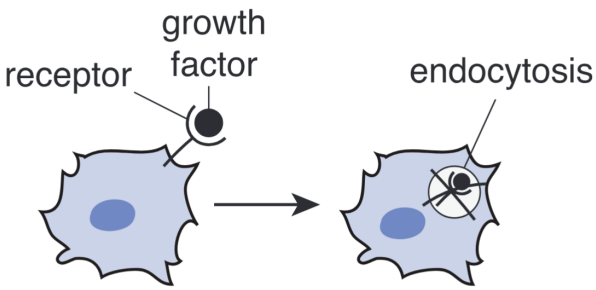



Figure 9: Endocytosis is the intake of molecules that bind a receptor into the cell resulting in their degradation.

The $F$-specific factor $c_1$ is secreted by both cell types and endocytosed by its receivers $F$ cells. The $M$-specific factor $c_2$ is secreted by $F$ and endocytosed by $M$ cells.

$$\frac{dc_1}{dt} = aF + b_1 M - e_1 F c_1 - \gamma_1 c_1 \tag{3.1}$$

$$\frac{dc_2}{dt} = b_2 F - e_2 M c_2 - \gamma_2 c_2 \tag{3.2}$$

Here $b_1$ is production rate per $M$ cell, $b_2$ is production rate per $F$ cell, and endocytosis rates like $e_1, e_2$ are about 1000/cell/min. Therefore, endocytosis is the main removal mechanism of growth factors unless cell density is very low. Growth factors dynamics have timescales of minutes to hours, whereas cell growth is much slower with a timescale of days. We thus invoke separation of timescales and compute the quasi-steady state of the two growth factor concentrations by setting $dc_1/dt = 0$ and $dc_2/dt = 0$. Balancing secretion with endocytosis and degradation gives

$$c_1 = \frac{aF + b_1 M}{e_1 F + \gamma_1} \qquad c_2 = \frac{b_2 F}{e_2 M + \gamma_2} \tag{3.3}$$

On the slower timescale of cell turnover, myofibroblasts divide under the control of $c_1$ and macrophages under the control of $c_2$. Macrophages are far from their carrying capacity, so the factor $1 - F/K$ enters only the $F$ equation

$$\frac{dF}{dt} = p_1 c_1 F \left(1 - \frac{F}{K}\right) - d_1 F \tag{3.4}$$

$$\frac{dM}{dt} = p_2 c_2 M - d_2 M \tag{3.5}$$

Substituting the quasi-steady-state growth-factor values for $c_1$ and $c_2$ gives

$$\frac{dF}{dt} = F\left[p_1 \frac{aF + b_1 M}{e_1 F + \gamma_1}\left(1 - \frac{F}{K}\right) - d_1\right] \tag{3.6}$$

$$\frac{dM}{dt} = M\left[p_2\frac{b_2F}{e_2M+\gamma_2} - d_2\right] \tag{3.7}$$

## 3.2 Nullclines and Phase Portrait

To understand these equations, we use the method of nullclines. Nullclines are curves in which one of the two cell concentrations does not change. One nullcline is $dM/dt = 0$, and the other is $dF/dt = 0$. The fixed points are where the two nullclines intersect. We Plot the nullclines on the phase plane, with $F$ on the horizontal axis and $M$ on the vertical axis, and consider the region $0 < F < K$ and $M > 0$. At each point in the plane, we plot a little arrow showing where $M$ and $F$ flow to if they start at that point. The arrows indicate the direction of flow.

The $dM/dt = 0$ nullcline gives the horizontal axis $M = 0$, and the solution to

$$p_2\frac{b_2F}{e_2M+\gamma_2} - d_2 = 0$$

Solving for $M$ gives

$$M = \frac{p_2b_2}{d_2e_2}F - \frac{\gamma_2}{e_2} \tag{3.8}$$

This is a straight line $M = \alpha F - \beta$, with slope $\alpha = p_2b_2/(d_2e_2)$ and an intercept $\beta = \gamma_2/e_2$. The intercept is close to zero because endocytosis dominates degradation and thus $\beta \ll 1$. The $dM/dt = 0$ nullcline separates the phase plane into two regions. At fixed $F$, the net growth rate per macrophage decreases as $M$ increases. Above the line, $dM/dt < 0$ and the arrows point down. Below the line, $dM/dt > 0$ and the arrows point up.

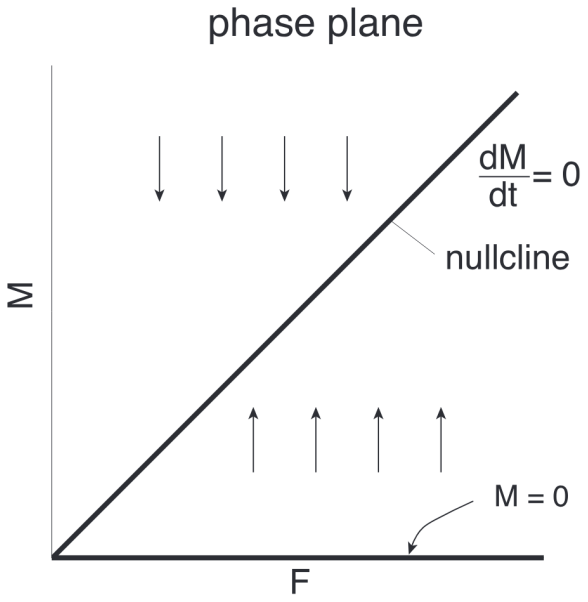



Figure 10: Each nullcline divides the phase plane into regions where the relevant variable moves in one direction.

The $dF/dt = 0$ nullcline gives the vertical axis $F = 0$, and the solution to

$$p_1 \frac{aF + b_1 M}{e_1 F + \gamma_1} \left(1 - \frac{F}{K}\right) - d_1 = 0$$

Solving for $M$ gives

$$M = \frac{d_1 e_1}{b_1 p_1} \left(\frac{F + \gamma_1/e_1}{1 - F/K}\right) - \frac{a}{b_1} F \tag{3.9}$$

We can understand the U-shape intuitively from the equation. After multiplying the nonzero-$F$ condition for $dF/dt = 0$ by $1 - F/K$, the autocrine-growth and carrying-capacity terms generate an $F^2$ term. Thus the nullcline is nonlinear rather than a straight line. At $F = 0$, Eq. (3.9) gives $M(0) = d_1\gamma_1/(b_1 p_1) > 0$: the branch starts above the origin. In the parameter regime shown here, the curve initially slopes downward. But as $F \to K^-$, the denominator $1 - F/K$ in Eq. (3.9) approaches zero and $M(F)$ rises without bound. The curve therefore turns upward, giving a U-shaped branch. The $F^2$ term signals the nonlinearity; the initial downward slope is also needed for the U-shape.

Plotting the $dF/dt = 0$ nullcline, we see that it has a U-shape which descends through the unstable fixed point $F_u$, drops below the horizontal axis and rises through the high fixed point $F_{\text{high}}$, and then climbs up and diverges near the carrying capacity $F = K$. At fixed $F$, increasing $M$ raises $dF/dt$. Therefore, above the curve, $dF/dt > 0$ and the arrows point right. Below the curve, $dF/dt < 0$ and the arrows point left.

| Nullcline | Above | Below |
|---|---|---|
| $M$-nullcline ($dM/dt = 0$) | $\downarrow$ | $\uparrow$ |
| $F$-nullcline ($dF/dt = 0$) | $\rightarrow$ | $\leftarrow$ |

We combine these two directions of motion to draw the arrows in the phase plane. On the $dF/dt = 0$ nullcline, only $M$ changes, the arrows are vertical. On the $dM/dt = 0$ nullcline, only $F$ changes, the arrows are horizontal. The nullclines intersect at fixed points, where neither cell population changes.

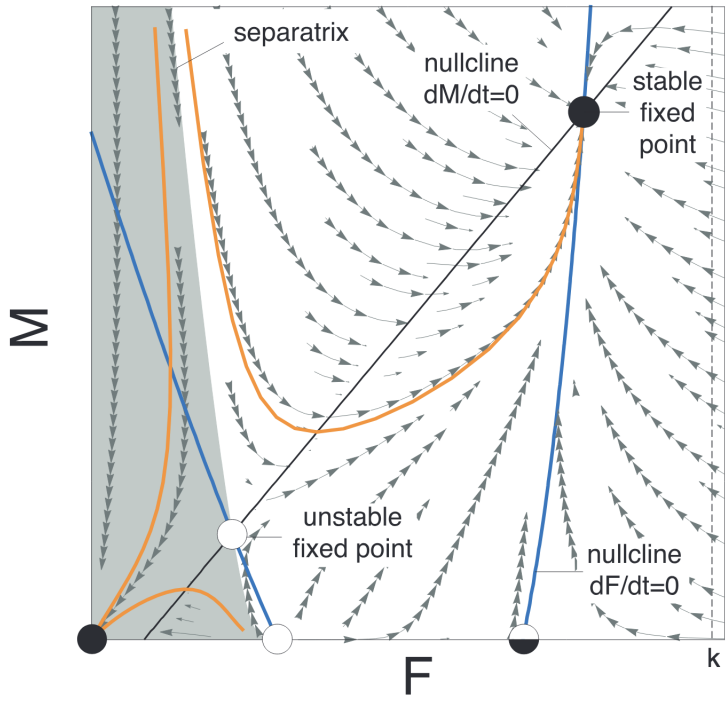



Figure 11: Nullclines and phase portrait for the fibroblast-macrophage circuit. Selected trajectories are shown in orange.

The phase portrait can be experimentally measured *in vitro*. Zhou et al. plated cells at many different initial concentrations, and watched how the cell concentrations changed over 2 days, plotting little arrows to indicate the changes. [2]

## 3.3 Fixed Points and Hot/Cold Fibrosis

The phase portrait reveals two stable fixed points. There is a fixed point at zero cells ($F = 0$, $M = 0$). This is the **OFF state** that corresponds to healing, since the myofibroblasts and macrophages are gone.

The other stable fixed point has high levels of both $M$ and $F$ cells, which sustain each other. This is the **ON state** that corresponds to fibrosis. All arrows in the vicinity flow to it. If a perturbation around the ON state occurs, say that a few extra $M$ arrive, they eat up their own growth factor and cell numbers drop back to steady state. This is how the endocytosis process can provide a systems-level effect of stabilizing fixed points. The carrying capacity for $F$ is also essential to stabilize the ON state. [3]

To understand fibrosis, we further need to consider the fibers, namely the **Extracellular matrix (ECM, or $E$)** deposited by myofibroblasts. Macrophages produce molecular scissors (matrix metalloproteinases, MMPs) that degrade and remodel the matrix $E$; regular tissue cells provide baseline removal. Thus, $E$ rises with $F$ and drops with $M$. At the OFF state, the fibers $E$ go to zero. At the ON state, the fibers $E$ reach a high steady-state concentration, continually made by $F$ and degraded by $M$. The fibrotic scar is a living tissue.

The phase portrait shows another interesting fixed point, the **ON/OFF state** with only

fibroblasts. This state can be called **"cold fibrosis"** where cold means no immune cells, $M = 0$. The ON state can be called **"hot fibrosis"**.

The cold fibrosis fixed point can be stable for certain ranges of the circuit parameters, such as weak secretion by $F$ cells of growth factors for $M$ cells. We can expect that cold fibrosis is "worse" because there is a lack of $M$ cells with their molecular scissors, and the ECM is more abundant and stiff than in the ON state. Such stiff and abundant ECM in cold fibrosis is found in end-stage liver cirrhosis and in fibrosis after heart attacks.

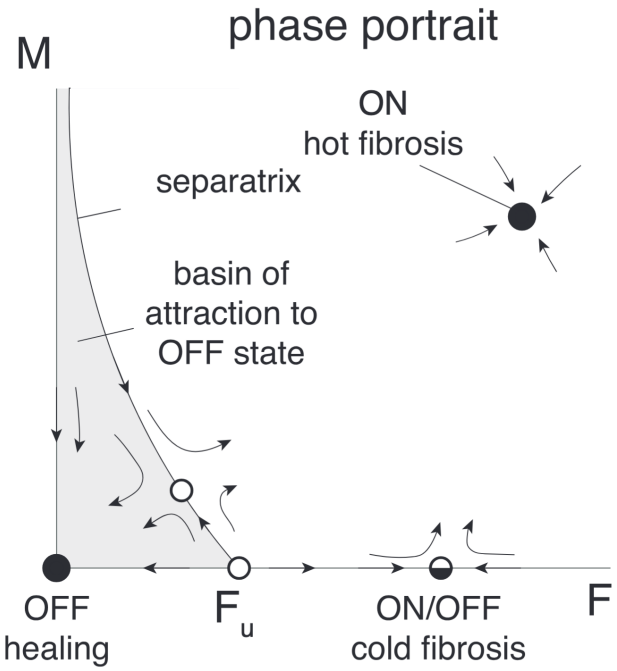



Figure 12: The phase portrait shows a stable ON state with both cell types, called hot fibrosis, a stable OFF state with neither, and a semi-stable state with only fibroblasts called cold fibrosis. A separatrix divides the phase portrait into basins of attraction for the two stable states, the ON and OFF states.

# 4. Injury and Inflammation Can Be Modeled by a Transient Influx of Macrophages

To see the dynamics of healing, let's consider an injury at $t = 0$, with a small initial number of $F$ and $M$ cells at the injury site. We model inflammation as a large influx $I(t)$ of macrophages. The influx adds cells directly to the tissue, so it enters $dM/dt$ as an additive term

$$\frac{dM}{dt} = M\left[p_2\frac{b_2F}{e_2M+\gamma_2} - d_2\right] + I(t) \tag{4.1}$$

Consider a 2-day pulse of inflammation in which influx $I(t)$ is high for 2 days and then returns to zero. A pulse raises $M$ sharply, and produce $c_1$. As a result, $F$ cells begin to divide. We again use the phase portrait but this time in log scale to show the region of low cell concentrations. [4]

**Transient injury**

A short pulse leaves the system within the basin of attraction to the OFF states (Fig. 13). The 2-day influx raises $M$ and $F$ temporarily, then both $M$ and $F$ levels decline until $F = 0$, $M = 0$ is reached. Scar fibers $E$ are deposited by $F$ cells, and when the $F$ cells are gone, scar is degraded by the tissue. Scar $E(t)$ rises and then vanishes. The timescale is about 2 weeks, using the typical parameters of Table 5.1.

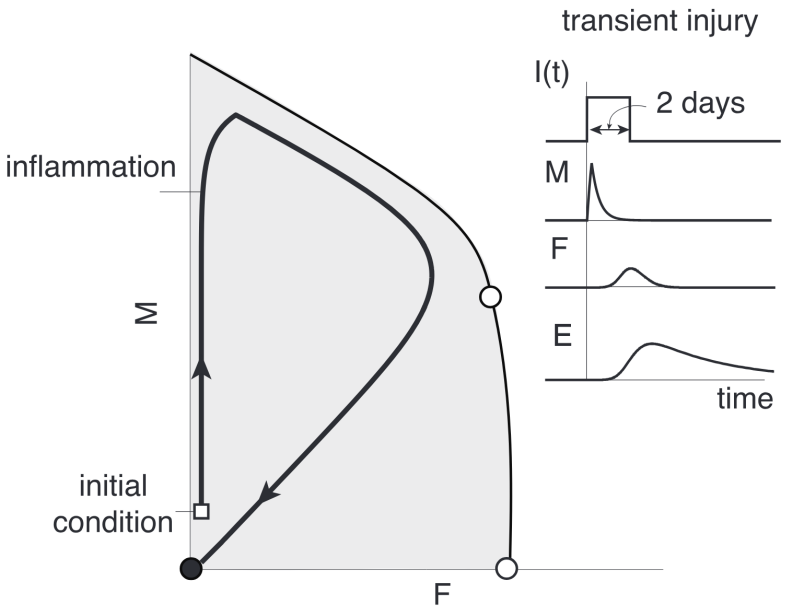



Figure 13: A transient injury causes an influx of macrophages and rise of fibroblasts but remains in the basin of attraction of the OFF state so that cell populations decline and the scar is removed.

### Prolonged injury

Now consider a longer pulse of inflammation which lasts for 4 days. $M$ levels rise sharply and cross the boundary called the **separatrix** to the basin of attraction to the ON state. Now there are enough $F$ and $M$ cells to support each other. The cells flow to the ON state. They create a scar tissue with constant turnover of $M$ and $F$ cells, and a high steady-state level of fibers $E$. Thus, a 4-day inflammation event leads to fibrosis.

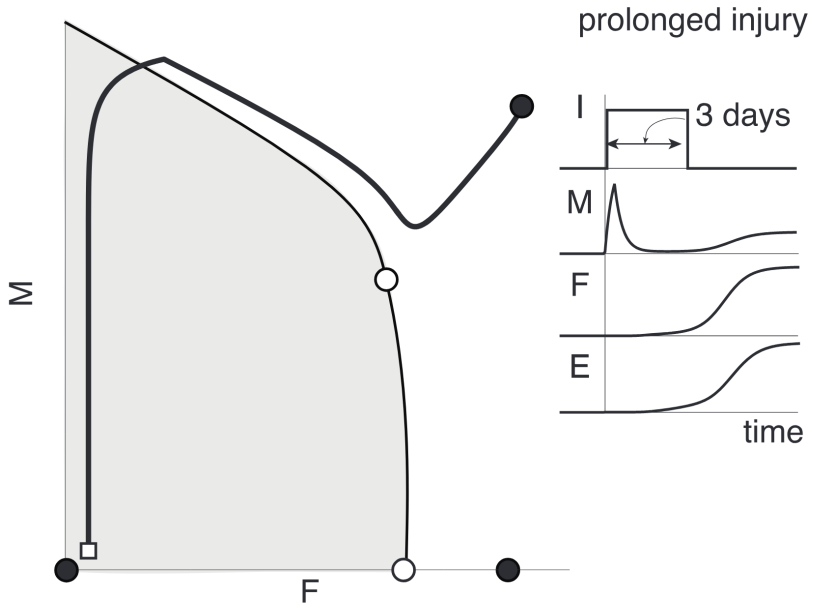



Figure 14: A prolonged injury crosses the separatrix into the basin of attraction of the ON state, causing fibrosis in which fibroblasts and macrophages support each other and continuously turn over.

### Repetitive injury

Similarly, consider a repeated injury. A 2-day inflammation pulse is not sufficient to cross the separatrix, but if another 2-day pulse occurs after a week, there are enough $M$ and $F$ cells left from the first injury to cross the separatrix and go to fibrosis.

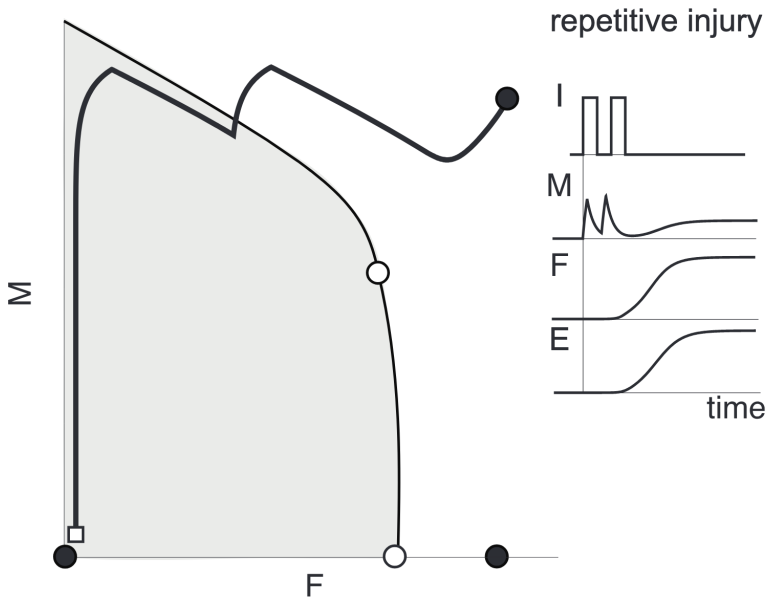



Figure 15: Recurring or repetitive injury can cross the separatrix and trigger fibrosis.

The following table gives representative parameter values for the Macrophage–Myofibroblast Cell Circuits.

**TABLE 5.1** Representative Parameter Values for Cell Circuits

| Parameter | Biological Meaning | Value |
|---|---|---|
| $p_1$ | Maximal proliferation rate of myofibroblasts | 0.9 $\text{day}^{-1}$ |
| $p_2$ | Maximal proliferation rate of macrophages | 0.8 $\text{day}^{-1}$ |
| $d_i$ | Removal rate of the cells | 0.3 $\text{day}^{-1}$ |
| $K$ | Carrying capacity of myofibroblasts | $10^6$ cells ( $\sim 10^{-3} \frac{\text{cell}}{\mu\text{m}^3}$ ) |
| $k_i$ | Binding affinity of growth factor $c_i$ | $6\times10^8$ molecules |
| $b_2$ | Maximal secretion rate of growth factor by myofibroblasts | $470 \frac{\text{molecules}}{\text{cell min}}$ |
| $b_1$ | Maximal secretion rate of growth factor by macrophages | $70 \frac{\text{molecules}}{\text{cell min}}$ |
| $a$ | Maximal autocrine secretion rate by myofibroblasts | $240 \frac{\text{molecules}}{\text{cell min}}$ |
| $e_2$ | Maximal endocytosis rate of growth rate by macrophages | $940 \frac{\text{molecules}}{\text{cell min}}$ |
| $e_1$ | Maximal endocytosis rate of growth rate by myofibroblasts | $510 \frac{\text{molecules}}{\text{cell min}}$ |
| $\gamma$ | Degradation rate of growth factors | 2 $\text{day}^{-1}$ |

*Source: Alon, Table 5.1 [1], adapted from Adler et al. [3].*

Note: This table uses the more accurate parameters for Michaelis–Menten functions for endocytosis and proliferation, which go as $c_i/(k_i + c_i)$ instead of linearly with $c_i$ as in the text.

# 5. The Time Window for Stopping Inflammation Is Due to Bistability

We can now understand why it is so urgent to stop inflammation to avoid fibrosis. Let us plot how the duration of the inflammation pulse $\Delta t$ affects the final steady-state amount of scar fibers $E_{\rm st}$.

Below a critical duration of inflammation, of about $t_c = 3$ days, the scar vanishes. Above $t_c$, $M$ crosses the separatrix and the ON state is inevitable even if the inflammation stops. For example, an inflammation pulse that lasts a bit longer than the critical threshold $t_c$ causes $M$ to cross the separatrix, and F cells have a bit of time to multiply. When inflammation stops, $M$ levels sharply drop, but recover due to the increased F cells. Both $F$ and $M$ go up together over weeks to the ON state. Fibrosis occurs even though the inflammatory pulse stopped long ago.

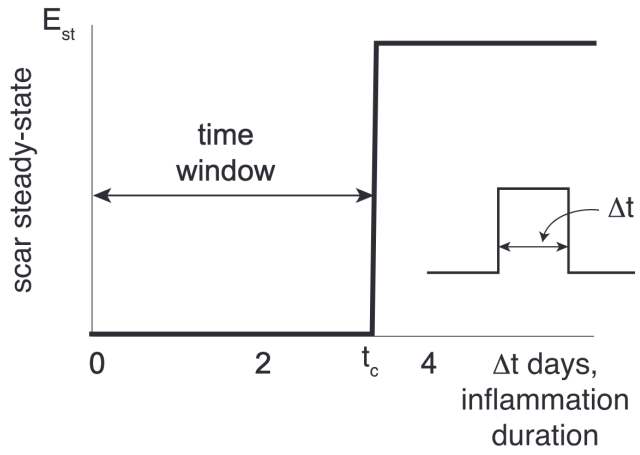



Figure 16: A time window for preventing fibrosis: inflammation lasting less than about 3 days avoids fibrosis whereas fibrosis is reached when inflammation lasts longer than 3 days.

# 6. The Long Timescale for Scar Maturation and Healing Is Due to the Slowdown Near an Unstable Fixed Point

Scar maturation is a process that unfolds over months – the scar changes until it reaches steady state. This timescale is much slower than the cell turnover time of days. How does the long timescale arise?

The slow timescale found in the model is due to the fact that the dynamics near the separatrix approach an unstable fixed point. By definition, at a fixed point, including an unstable one, the velocity is zero. Thus, the velocity is always slow near a fixed point, causing a slowdown phenomenon. The same applies to the healing process, which dawdles around the unstable fixed point and takes about 2 weeks to resolve back to the OFF state.

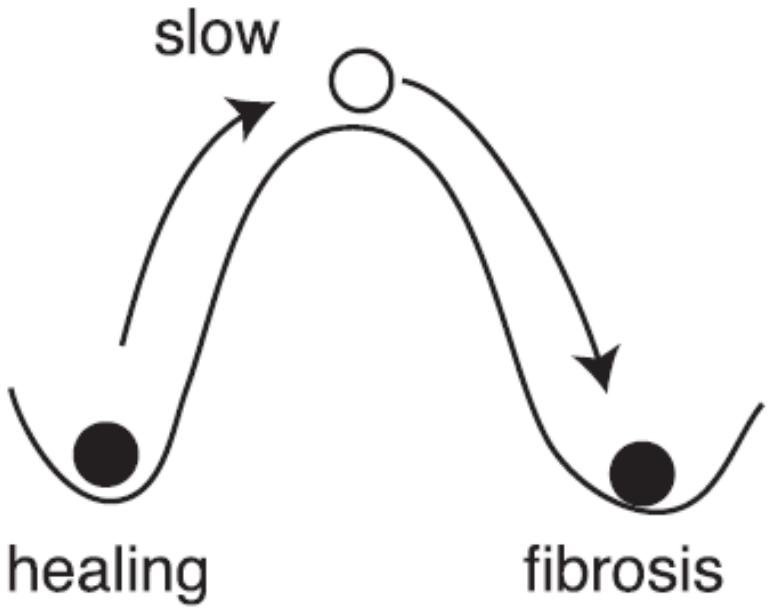



Figure 17: Dynamics are slow near an unstable fixed point, in analogy to a ball rolling to the top of a hill.

# 7. Strategies for Preventing and Reversing Fibrosis

To prevent fibrosis, we need to enlarge the basin of attraction for the OFF state. A large basin means that more situations end up resolved without fibrosis. We can also explore whether fibrosis can be reversed: can a mature scar in the ON state be made to flow to the OFF state?

It is found that prevention and reversal of fibrosis become possible in the model when the ON/OFF fixed point vanishes, because this greatly expands the basin of attraction of the OFF state. A key model strategy is to weaken the capacity of fibroblasts to sustain themselves through the autocrine loop. In the simplified one-cell model, eliminating the positive fixed points requires

$$\frac{p_1 aK}{\gamma_1 d_1} < 4$$

When we reduce this parameter group, the unstable fixed point on the $x$ axis moves to higher and higher values, enlarging the basin of attraction to the OFF state. At a certain value, the unstable fixed point collides with the ON/OFF fixed point, and both fixed points annihilate. The basin of attraction is now very large.

This parameter group offers several targets against fibrosis. One target suggested by the circuit equation is to inhibit the autocrine growth factor for myofibroblasts. This can be achieved by increasing its removal rate $\gamma$ or reducing its production rate $a$. Both changes push the parameter group down.

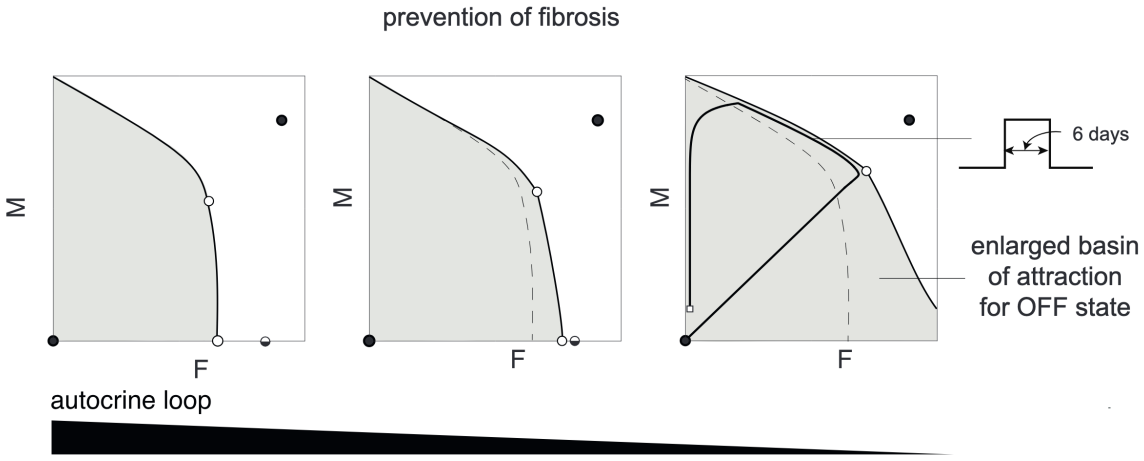



Figure 18: Strategy to resolve fibrosis by inhibiting the autocrine loop of fibroblasts. This enlarges the basin of attraction to the healing state.

# References


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# Terminology

| English | 中文 |
|---|---|
| Inflammation | 炎症 |
| Rubor (redness) | 红（发红） |
| Tumor (swelling) | 肿（肿胀） |
| Calor (heat) | 热（发热） |
| Dolor (pain) | 痛（疼痛） |
| Fibrosis | 纤维化 |
| Hot fibrosis | 热纤维化 |
| Cold fibrosis | 冷纤维化 |
| Tissue repair | 组织修复 |
| Scar maturation | 瘢痕成熟 |
| Innate immune system | 先天免疫系统 |
| Monocyte | 单核细胞 |
| Macrophage | 巨噬细胞 |
| Fibroblast | 成纤维细胞 |
| Myofibroblast | 肌成纤维细胞 |
| Epithelial cell | 上皮细胞 |
| Endothelial cells | 内皮细胞 |
| Senescent cell | 衰老细胞 |
| Apoptosis | 细胞凋亡 |
| Growth factor | 生长因子 |
| Autocrine loop | 自分泌回路 |
| Endocytosis | 内吞作用 |
| Homeostasis | 内稳态 |
| Extracellular matrix (ECM) | 细胞外基质 |
| Matrix metalloproteinase (MMP) | 基质金属蛋白酶 |
| Tissue inhibitor of metalloproteinases (TIMP) | 金属蛋白酶组织抑制因子 |
| Cell turnover | 细胞更新 |
| Matrix remodeling | 基质重塑 |
| Carrying capacity | 承载量（细胞密度上限） |
| Logistic growth | 逻辑斯蒂增长 |
| Bistability | 双稳态 |
| Multistability | 多稳态 |
| Basin of attraction | 吸引域 |
| Separatrix | 分界线（吸引域边界） |
| Nullcline | 零流线 |